What was tried before the multi-regime frontier preflight, why, what the data showed, and what remained open

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Executive result. Across the prior ePC programme, the most reproducible positive finding was monotone predictive-coding activity energy under the implemented backtracking rule. The recurrent negative pattern was that this structural property did not translate into better held-out loss, adaptation, retained factor information, or stable conditional specialization than matched ordinary controls. The experiments nevertheless established useful plumbing and positive controls. The multi-regime preflight was attempted because the existing tiny fixture was too simple to rule out the possibility that a richer task might reveal viable, complementary ePC variants; it then failed its ordinary-control learnability gate.

1. Scope and how to read this history

This is not a meta-analysis of a single homogeneous benchmark. The underlying work comprises five distinct question families:

1. Can block-state ePC be implemented faithfully enough that T=1 is the ordinary KD endpoint and T>1 has well-defined, monotone activity inference?
2. On a six-layer GPT-2 student, does ePC improve source-domain distillation or downstream adaptation relative to matched backpropagation?
3. Does ePC produce qualitatively useful representation geometry, robustness, or causal-factor retention?
4. Can semantic-free winner assignment maintain multiple coherent continuations, and does ePC improve that mechanism?
5. Does the ePC configuration frontier contain several *diversely good* candidates with stable conditional crossover, sufficient to motivate a later MoE?

Negative results at one level do not logically refute the whole method. Conversely, a positive structural invariant or toy positive control does not establish a language-scale benefit. Each section below separates **observed** measurements from inference.

2. The tested ePC mechanism

The transformer is treated as a chain of residual-block predictions. With frozen weights, input state z_0, block states z_l, block prediction f_l, and teacher distribution q, the implemented local energy was

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E(z) = 1/2 Σ_l mean ||z_l − f_l(z_{l−1})||² + λ KL(q || softmax(logits(z_L)/τ)).
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T=1 is exactly the ordinary knowledge-distillation endpoint. At T>1, residual-stream activities are relaxed using gradients of this declared energy. A backtracking line search accepts a step only if the energy does not increase; the settled states are detached, and the local energy is differentiated with respect to weights. Thus energy monotonicity is a check on activity inference, not a claim that training loss decreases monotonically.

The repeatedly used minimal invariants were finite values/gradients, exact replay from matched initial states and RNG, and monotone activity energy for ePC arms. The important outcome metrics varied by experiment: validation loss/perplexity, adaptation AUC, forgetting, effective rank, causal selectivity, planted-mode coverage, and conditional regime NLL.

3. Chronology at a glance

Table 1. Major prior experiments relevant to the later frontier question.

Date / ledger	Question and scale	Key result	What it changed
15 Jul `epc-distillation-gate-v2`	Three-seed local correctness/credit gate.	Normalization restored earlier-block credit; no ePC depth beat both controls.	Established real local-credit plumbing but blocked promotion.
17–21 Jul r7, r8, r9	6-layer GPT-2, WikiText-103, 1 then 5 seeds.	Energy monotone 10/10 in R8; ePC worse than BP+CE at matched updates and slower.	Shifted interest from raw loss to representation and conditional outcomes.
18–19 Jul 6-layer outcome battery	Frozen-backbone WikiText→TinyStories adaptation, 3 seeds.	ePC had worse target AUC than both BP controls, despite less source forgetting and distinct low-rank geometry.	Different representations did not supply functional benefit.
1 Aug exploration/balancing	Local planted two-mode transformers, 3-seed gates.	Winner assignment and context balancing can recover modes; T=4 did not robustly beat T=1.	Created a semantic-free testbed for specialization, with strict conditional coverage.
3 Aug Pareto v1/v2	14 ePC settings, exploration/selection	Conditional variation existed but no candidate survived	No MoE on this fixture; propose a materially richer

/confirmation splits. confirmation viability. task.

4. Early implementation and credit-assignment gates

4.1 Why these gates were necessary

Early ePC failures could have reflected a trivial implementation error: a local energy term can be correctly decreasing while sending essentially no credit to early blocks. Before interpreting performance, the work therefore tested whether the declared T=1 endpoint was equivalent to ordinary KD, whether energy was monotone, and whether local-error normalization restored nonzero early-block gradients.

4.2 Corrected local-credit gate (15 July)

Three untouched seeds (73, 89, 107) compared ordinary BP, KD, and ePC depths 2/4/8. At λ=0.05 on seed 73, the block-1 ePC/KD gradient-norm ratio rose from about 10^{-4} under the earlier formulation to 0.050, 0.136, and 0.265 at T=2/4/8. Block-0 credit became nonzero at T=4 (0.0071) and T=8 (0.0404); energy and finite-gradient invariants passed.

But the performance promotion gate failed. Mean perplexity was BP 75.1650; KD 75.1622, 75.1405, 75.1299 across its settings; and ePC 75.1577, 75.1551, 75.1503 for T=2/4/8. The corrected ePC machinery therefore produced credit but did not beat ordinary KD, and took substantially longer (at the reported setting, roughly 0.97 s versus 0.033 s for KD). **Inference:** a real normalization bug was fixed, but corrected credit alone was not a performance mechanism.

5. Six-layer GPT-2 distillation and representation studies

5.1 Single-seed mechanism screen r7

R7 was the completed repair of a previously crashing five-arm screen. It used a 6-layer GPT-2-width student (d=768, 12 heads, d_{ff}=3072) and 12-layer GPT-2 teacher on 200 WikiText-103 documents, 64 eval segments, context length 256, 200 updates, effective batch 32, AdamW 1e−4, bf16, KD temperature 2.0, and seed 3253. Arms were BP+CE, update-matched BP+KD, ePC T=4, ePC T=12, and wall-clock-matched BP+KD.

Table 2. R7 final validation loss, single seed 3253. This is diagnostic, not a confirmation study.

BP+CE	BP+KD	ePC T=4	ePC T=12	wall-clock BP+KD
8.4390	7.6278	7.5672	7.5836	7.6278

The ePC arms were monotone and slightly below update-matched KD in this one seed, but T=12 did not improve over T=4 while costing about four times more. This was explicitly insufficient for a performance conclusion.

5.2 Five-seed R8 broad outcome screen

R8 expanded to seeds 1729, 3253, 6421, 8191, and 10103; 25 records (five seeds × five arms) completed on a pinned configuration. The teacher reference was validation loss 4.124 and perplexity 61.8. All ten ePC records (five seeds × T=4/T=12) had monotone activity energy.

Table 3. R8 source validation outcomes, mean ± SD across five seeds.

Arm	Val loss	Val PPL	KD gap	Updates	Elapsed s
BP+CE	6.261 ± .060	524.3	1160.9	200	78
BP+KD	7.039 ± .056	1141.6	823.8	200	74
ePC T=4	6.781 ± .146	888.6	978.0	200	445
ePC T=12	6.754 ± .131	863.2	960.6	200	1738
BP+KD wall-clock	5.785 ± .135	327.5	469.8	1334	445

Observed: at 200 updates BP+CE beat both ePC depths on loss. At the ePC-T=4 wall-clock budget, ordinary KD took 6.7× more updates and dominated all arms on loss. T=12 was marginally better than T=4 but well within its seed-level variation and much slower. The strongest positive result remained the 10/10 energy-monotonicity check.

5.3 R9 representation/cause probes

R9 analyzed the R8 checkpoint family. At the final layer, BP+CE had entropy effective rank 13.9±0.3 and participation ratio 6.1±0.1; BP+KD had 5.0±0.1 and 3.1±0.0; ePC T=4 had 3.9±0.2 and 2.6±0.2; ePC T=12 had 3.8±0.1 and 2.6±0.1. Dead fraction was 0.001 throughout. Thus ePC did not create sparse dead units; it concentrated active variation into a much lower-dimensional subspace.

Table 4. R9 final-layer causal selectivity, mean ± SD across five seeds.

Arm	Subject number	Object number	Tense	Negation
BP+CE	.607±.017	.425±.044	.661±.054	.870±.013
BP+KD	.514±.030	.302±.068	.592±.034	.720±.021
ePC T=4	.438±.065	.232±.035	.546±.028	.719±.044
ePC T=12	.379±.072	.275±.026	.481±.066	.718±.052

Inference: the measured energy smoothness is associated here with compression and weaker factor selectivity, not an obviously useful factorization. It remains possible that a different objective or capacity would turn compression into a benefit; the reported configuration did not.

6. Functional adaptation battery: different geometry was not enough

A separate frozen six-layer outcome battery tested whether ePC checkpoints had a better plasticity/forgetting tradeoff. Three seeds (1729/3253/6421) produced BP+CE, BP+KD, and ePC+KD checkpoints. For each checkpoint, the backbone was frozen, rank-8 low-rank adapters (α=16) were fitted to WikiText→TinyStories under the same 100-update rule. The primary endpoint was target validation-loss AUC (lower is better); promotion also constrained initial target regression and source forgetting.

Table 5. Adaptation evidence (per-seed means).

Seed	Arm	Pre-target loss	Source forgetting	Target gain	Target AUC
1729	BP+CE / BP+KD / ePC	7.895 / 5.835 / 6.908	.257 / .197 / .114	2.198 / .962 / .645	6.351 / 5.169 / 6.442
3253	BP+CE / BP+KD / ePC	7.735 / 5.828 / 6.994	.331 / .149 / .064	2.075 / .961 / .714	6.286 / 5.162 / 6.476
6421	BP+CE / BP+KD / ePC	7.786 / 5.887 / 7.086	.239 / .134 / −.040	2.132 / 1.023 / .773	6.283 / 5.168 / 6.497

Bootstrap control-AUC-minus-ePC-AUC contrasts were −0.1645 (95% CI −.2212, −.0949) against BP+CE and −1.3043 (−1.3336, −1.2697) against BP+KD; positive would have favored ePC. ePC mean source forgetting was only .0461, but its initial target-loss regressions and poorer target AUC violated the frozen promotion rule. CKA and rank probes showed ePC geometry was very different (mean effective rank about 4.0–4.8 versus 55–57 for BP+CE), but no functional advantage followed.

7. Semantic-free exploration: what did work on a toy fixture

7.1 Known-answer winner-assignment positive control

The programme then isolated a narrower question: can a learner preserve two coherent alternative continuations rather than their invalid marginal average? On a planted two-head/two-mode fixture, seeds 1709/3253/6421 compared marginal T=1, marginal T=4, explorative T=1, and explorative T=4. All twelve records replayed exactly; all ePC traces were monotone.

Mean planted prototype-coverage MSE was .999498 (marginal T=1), .999496 (marginal T=4), .000227159 (explorative T=1), and .000226819 (explorative T=4). Both explorative arms recovered both modes on all seeds, with effective mode count 1.998632. This was a genuine positive control for winner-assigned multi-continuation training, but the T=4 advantage over T=1 was only 3.40×10^{-7} MSE—no evidence for ePC superiority.

7.2 Transformer transfer and conditional failure

On a tiny causal transformer with two contexts, pure winner-take-all substantially improved coverage NLL from .72199/.71975 (marginal T=1/T=4) to .28580/.30306 (explorative T=1/T=4). Yet it recovered both modes in both contexts only on seed 6421; seeds 1709 and 3253 lost a conditional mode. Global assignment entropy looked healthy even for collapsed heads, proving it was not a sufficient diversity metric.

Context-balanced capacity assignment repaired that structural defect on fresh seeds 7817/9151/12011: both balanced arms recovered both modes in both contexts on all 3/3 seeds. Balanced T=1 coverage NLL was .2373, .2396, .2247, passing the <.25 gate. Balanced T=4 was .3412, .2322, .2182: it missed one seed. A fresh calibration grid over T∈{2,4,6} and relaxation {.1,.2} then found no all-seed ePC pass; closest T=6/.2 had mean .2577, and even the contemporaneous balanced T=1 control missed .2689/.3077/.3085 on fresh seeds. The result was seed-sensitive mechanism behavior, not a validated ePC gain.

8. The explicit diversity/Pareto experiment

The MoE-motivated question was made explicit in the 3 August frontier study. A viable ePC setting needed exact replay, monotone energy, both modes in both contexts, coverage NLL below an absolute floor, and no excessive loss versus the matched T=1 control. A *diversely good* pair additionally needed repeatable bidirectional regime wins on untouched confirmation seeds.

V1 screened 14 configurations across six controls. On exploration seeds 31001/31002, four configurations were viable (`out05`, `t6_r02`, `temp125`, `upd128`), but aggregate Pareto axes retained only `t6_r02`. Since a conditional specialist might be pruned by aggregate objectives, V1 was treated as an audit finding, not the answer.

V2 repaired the objective by retaining every context×mode NLL as a Pareto axis and used disjoint seeds: exploration 34001/34002, selection 35001/35002, confirmation 36001/36002/36003. Exploration retained `t6_r02` (coverage NLL .1874, effective rank 9.62) and `upd128` (.1803, rank 6.26). Selection retained only `upd128` (.1283). On confirmation it scored .1950, .3213, .1426; seed 36002 recovered both modes in only one of two contexts. Thus zero candidates, and hence no pair, survived confirmation. All 62 V1/V2 records replayed exactly and were energy-monotone.

Conclusion of the prior frontier study. There was conditional variation, but it was seed-sensitive rather than a stable population of viable specialists. Building an MoE from that candidate pool was therefore rejected. This is a negative result for the balanced tiny-transformer fixture, not a general ePC impossibility theorem.

9. Why the multi-regime preflight followed

The earlier tiny fixture was deliberately suitable for checking mechanics, but it had only two contexts, two modes, short fixed continuations, and an explicit balancing pressure. Once a model learned both continuations, it offered little room for distinct ePC settings to have genuinely different functional niches. At the same time, six-layer language experiments showed that merely scaling ePC did not solve the problem: ePC was slower, lower-rank, and less selective in the tested setup.

The proposed bridge was therefore not “make the network bigger” alone. It was a larger, controlled task with independently measured in-distribution continuation, composition, perturbation/OOD, sequential adaptation, and retention, plus the same strict conditional-frontier logic. The companion report `failed_experiment_report.pdf` documents that preflight. Its matched T=1 control failed every final learnability floor; therefore even this successor did not reach the stage where an ePC frontier could be interpreted.

10. Current evidence and honest next constraints

- **Reproduced positive:** the implemented ePC activity inference can be deterministic and energy-monotone across multiple seeds and settings.
- **Reproduced negative:** in the tested six-layer GPT-2 configurations, ePC did not beat BP controls on held-out source loss, wall-clock efficiency, adaptation AUC, rank, or causal selectivity.
- **Reproduced positive control:** winner assignment plus context balancing can preserve planted multimodality without semantic labels; this does not distinguish T=1 from ePC.
- **Not demonstrated:** a stable diverse ePC frontier, a viable expert pool, an MoE benefit, a language-scale ePC advantage, or a general refutation of predictive coding.

A future successor must be prospectively specified with a task that a matched T=1 control demonstrably learns on fresh calibration seeds. Only then should it open new exploration/selection/confirmation splits. It should preserve conditional objectives separately, include matched wall-clock baselines, and treat any ePC advantage as provisional until it survives untouched confirmation.

11. Primary provenance

Authoritative ledgers: `20260715T154938Z-epc-distillation-gate-v2`; `20260720T114200Z-epc-mechanism-screen-r7`; `20260720T193211Z-epc-r8-broad-outcome`; `20260721T222400Z-r9-stage-a`; `20260801T151100Z-explorative-epc-baseline`; `20260801T154000Z-tiny-transformer-explorative-epc`; `20260801T160000Z-tiny-transformer-balanced-exploration-v2`; `20260801T160747Z-balanced-epc-calibration-v1`; `20260804T055941Z-epc-diverse-pareto-frontier-v1`; and `20260804T060519Z-epc-diverse-pareto-frontier-v2`, all under `projects/relaleap/experiments/`. Detailed six-layer adaptation evidence is in `docs/relaleap_gpt2_six_layer_outcome_report.pdf`. The historical report itself is a synthesis; the run ledgers and hashed raw artifacts remain authoritative for every numerical claim.

Epistemic labels: reported numerical measurements are observed from recorded ledgers; comparisons across different studies are contextual, not pooled statistical estimates; explanations of compression, interference, or task insufficiency are hypotheses unless explicitly measured.